

MBCIE – Centre for AI

Technical Assessment — Agentic AI Data Science Assignment

Candidate: Siddharth

Duration: 2 Days from receipt

Dataset

Olist Brazilian E-Commerce Public Dataset (Kaggle): kaggle.com/datasets/olistbr/brazilian-ecommerce

Objective

Build an Agentic AI Business Analyst that can independently analyze the Olist dataset and answer open-ended business questions. Given a question, the agent should be able to:

- Understand the user's question and decide which data and tools are required
- Query and analyze the dataset; perform calculations and/or ML analysis
- Identify anomalies and trends in the data
- Generate business insights and actionable recommendations
- Clearly explain the reasoning and evidence behind every answer

Example Scenario

User asks: “Why did sales decline, and which sellers contributed most to it?”

Expected agent flow: Understand → Query Data → Analyze → Compare → Identify Causes → Recommend Actions

Minimum Requirements

1. Agentic workflow with multiple tools/functions
2. SQL / data-analysis tool
3. At least one ML capability (e.g., forecasting, clustering, anomaly detection)
4. Business analytics dashboard or user interface
5. Memory/state to maintain conversation context
6. Grounded answers based on actual dataset results (no hallucinated figures)
7. Error handling and validation
8. Architecture diagram and documentation

Deliverables

- Working application
- Source code / GitHub repository with README
- Architecture diagram
- 5–10 minute recorded demo walkthrough

Bonus

Implement multiple specialized agents — e.g., Data Analyst, ML Analyst, and Business Advisor — coordinated by a Manager Agent.

Evaluation Criteria

Agent autonomy • Tool usage • Reasoning workflow • Business value • Production-quality implementation

Submit your repository link and demo to Kumares.h.r@mbcie.org within the deadline. | MBCIE – Centre for AI, Ludhiana